

1. Executive Summary

Paracetamol and ibuprofen are the two most widely used over-the-counter (OTC) analgesic-antipyretic agents worldwide. Both provide relief of mild-to-moderate pain and fever, yet they differ markedly in pharmacology, safety profile, and therapeutic niche.

Feature	Paracetamol	Ibuprofen
Class	Analgesic-antipyretic (non-NSAID)	Non-steroidal anti-inflammatory drug (NSAID)
Key Mechanism	Central COX-3 inhibition → ↓ CNS prostaglandins (minimal peripheral anti-inflammatory effect)	Reversible inhibition of COX-1/COX-2 → ↓ peripheral prostaglandins (analgesic, antipyretic, anti-inflammatory)
Typical Adult OTC Dose	500 mg–1 g q4–6 h (max 4 g/24 h)	200 mg–400 mg q6–8 h (max 1.2 g/24 h OTC; 2.4 g/24 h prescription)
Primary Indications	Pain (headache, dental, musculoskeletal, dysmenorrhea) and fever	Same + conditions with an inflammatory component (arthritis, sprains, bursitis)
Major Safety Concerns	Dose-dependent hepatotoxicity; rare severe skin reactions	GI ulceration/bleeding, renal impairment, hypertension, cardiovascular risk, NSAID-sensitive asthma
Contra-indications	Severe hepatic disease, hypersensitivity, chronic alcohol >3 drinks/day	Active GI ulcer/bleed, severe renal/hepatic impairment, uncontrolled hypertension, NSAID hypersensitivity
Regulatory Status (US)	OTC monograph; prescription IV formulation (approved 2010)	OTC monograph; prescription higher-strength tablets (approved 1995)
Preferred Clinical Scenarios	Fever in children, patients with renal or cardiovascular risk, pregnancy ≥ 30 wk, aspirin-sensitive asthma	Inflammatory pain, musculoskeletal conditions, when anti-inflammatory effect is required, short-term postoperative pain

Overall, paracetamol is the first-line agent for isolated pain or fever when safety concerns limit NSAID use, whereas ibuprofen is preferred when an anti-inflammatory effect is clinically important and the patient has no contraindicating GI, renal, or cardiovascular risk factors.

2. Research Findings

The comparative data presented are drawn exclusively from the supplied source material, which collates FDA labeling, EMA public assessments, WHO essential-medicine listings, and pharmacovigilance observations. No external studies or unpublished data were introduced.

Key points extracted:

1. Pharmacokinetics – Both agents are rapidly absorbed (peak 1–2 h). Paracetamol undergoes hepatic glucuronidation/sulfation (minor CYP2E1) and is eliminated renally as metabolites. Ibuprofen is metabolised primarily by CYP2C9/2C19 and excreted unchanged and as metabolites via the kidneys.

2. Safety Signals –

- Paracetamol: hepatotoxicity is dose-dependent; risk amplified by chronic alcohol use, fasting, or CYP inducers.
- Ibuprofen: gastrointestinal irritation/ulceration, renal impairment (especially in volume-depleted, elderly, or CKD patients), hypertension, and potential cardiovascular events.

3. Drug-Drug Interactions – Summarised in Section 4 (see Table 4.1).

4. Regulatory Labels – FDA requires explicit warnings for severe liver injury with paracetamol (≥ 4 g/24 h) and for GI bleeding, cardiovascular risk, renal impairment, and pregnancy contraindication after 30 weeks gestation with ibuprofen.

No new clinical trial data were supplied; therefore, the report reflects the consensus of existing regulatory and pharmacovigilance information.

3. Medical Analysis

3.1 Therapeutic Uses

Clinical Situation	Paracetamol	Ibuprofen
Fever (all ages)	First-line; safe in children	Acceptable after 6 months, but paracetamol preferred in dehydration or renal risk
Non-inflammatory pain (headache, dental, mild musculoskeletal)	Effective; no GI/renal impact	Effective; may be chosen if additional anti-inflammatory effect desired
Inflammatory pain (arthritis, sprain, bursitis)	Limited benefit	Preferred – provides anti-inflammatory action
Pregnancy (≥ 30 weeks)	Preferred; category B/US	Contra-indicated (risk of premature ductus arteriosus closure)
Aspirin-exacerbated respiratory disease (AERD)	Preferred – does not trigger bronchospasm	May precipitate asthma exacerbation
Patients with chronic liver disease	Avoid – hepatotoxicity risk	Acceptable if renal function adequate
Patients with chronic kidney disease or heart failure	Preferred – minimal renal impact	Use with caution; risk of renal perfusion decline
Concurrent anticoagulation (warfarin)	May increase INR at high doses (> 3 g/day)	Increases bleeding risk via platelet inhibition

3.2 Advantages & Limitations

Attribute	Paracetamol	Ibuprofen
Platelet function	No effect – safe with anticoagulants (within dose)	Inhib